

Amphiphilic Zwitterionic Acrylate/Methacrylate Copolymers for Marine Fouling-Release Coatings

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Cite This: *Langmuir* 2021, 37, 5591–5600

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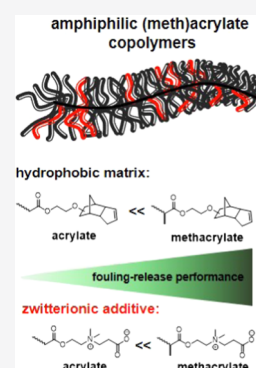
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ABSTRACT: Methacrylate and acrylate monomers are popular building blocks for antifouling (AF) and fouling-release (FR) coatings to counteract marine biofouling. They are used in various combinations and often combined into amphiphilic materials. This study investigated the FR properties of amphiphilic ethylene glycol dicyclopentenyl ether acrylate (DCPEA) and the corresponding methacrylate (DCPEMA) blended with 5 wt % zwitterionic carboxybetaine acrylate (CBA) and the corresponding methacrylate (CBMA). A series of (co)polymers with different acrylate/methacrylate compositions were synthesized and tested against the attachment of the diatom *Navicula perminuta* and in short-term dynamic field exposure experiments. The more hydrophobic methacrylate DCPEMA homopolymer outperformed its acrylate counterpart DCPEA. Incorporated zwitterionic functionality of both CBMA and CBA imparted ultralow fouling capability in the amphiphilic polymers toward diatom attachment, whereas in the real ocean environment, only the employment of CBMA reduced marine biofouling. Moreover, it was observed that CBA-containing coatings showed different surface morphologies and roughnesses compared to the CBMA analogues. Particularly, a high impact was found when acrylic CBA was mixed with methacrylic DCPEMA. While the wettability of the coatings was comparable, investigated methacrylates in general exhibited superior fouling resistance compared to the acrylates.



INTRODUCTION

The detrimental effects of biofouling in the shipping industry, which is the undesired accumulation of biomass on artificial submerged surfaces such as ship hulls, have been successfully counteracted by biocidal ablative paints.^{1,2} As the application of many of the used biocides including tributyltin (TBT) is increasingly restricted, the research on environmentally friendly alternatives has gained increasing attention.³ A myriad of studies on self-assembled monolayers (SAMs) provided profound knowledge on key surface properties for the design of effective coatings.^{4–7} To fully exploit the resistant properties of specific chemical building blocks, materials with the required physiochemical properties and a sufficient thickness of the coatings have to be developed and tested. Some of the prominent polymer classes are methacrylates and acrylates, as they can easily be converted into functional polymers by free radical polymerization. Once equipped with appropriate functional groups, acrylates and methacrylates can easily be applied in low fouling coatings.^{8–16} Among interesting classes of protective methacrylate coatings are poly(ethylene glycol) containing polymers.^{17–19} In general, it was noted that the thickness of the coatings (10–60 nm), e.g., hydroxy-functionalized methacrylate coatings, can influence their antifouling (AF) performance against proteins.¹⁵ Also, zwitterionic methacrylate polymers received major attention in recent years.^{20–26} For instance, on microporous membrane surfaces, sulfobetaine methacrylate (SBMA) coatings showed excellent

antifouling properties.¹⁶ Generally, the efficient use of zwitterionic methacrylates to prevent cell adhesion, the formation of biofilm, and algal attachment, has been intensively investigated.^{9–11,27} Other interesting chemistries counteracting the attachment of marine bacteria and diatoms are poly(ester-co-acrylates) functionalized with antifoulants, such as benzothiazolinone.¹² Also, UV-curable polyurethane acrylates blended with perfluoroalkyl acrylates revealed interesting properties for potential AF applications.¹³ When applied in a combinatorial way as polymer arrays, 22 acrylates and methacrylates were tested towards the accumulation of pathogenic freshwater bacteria, and several interesting polymers, such as ethylene glycol dicyclopentenyl ether acrylate (DCPEA), with very low fouling properties were discovered.¹⁴ In many cases, acrylates and methacrylates are copolymerized into amphiphilic polymers.^{28–31} Poly(ethylene glycol)-fluoroalkyl acrylates copolymerized with, for example, poly(dimethyl siloxane) methacrylates effectively reduce the settlement of barnacle cyprids and facilitate the removal of sporelings of the green algae *Ulva linza*.³¹ Carboxybetaine

Received: February 11, 2021

Revised: April 12, 2021

Published: April 30, 2021



methacrylate (CBMA) embedded in DCPEA were effective amphiphilic polymers to counteract attachment by *Navicula perminuta* and other marine organisms.³² This set of amphiphilic copolymers represents an interesting class of materials for prospective fouling-release (FR) coatings and raised the question if the performance can be fine-tuned further by adjusting the reactive group of the monomers.

If desired to combine zwitterionic carboxybetaine chemistry with dicyclopentenyl, there is a choice of either methacrylate or acrylate moieties. However, several differences in mechanical properties and reactivity between methacrylates and acrylates are well-documented.^{33,34} In general, methacrylates result in more rigid, hard coatings with higher tensile strength and higher glass transition temperature, while analogous acrylates form soft, rubbery, and more flexible coatings.³³ These contrasting properties are exploited in acrylate/methacrylate block copolymers such as thermoplastic elastomers or specialty adhesives, as they exhibit unique morphological, mechanical, and phase properties.³⁵ Usually highly defined block copolymers with blocks consisting of pure homopolymers are difficult to synthesize by conventional free radical polymerization.^{35,36} However, gradient copolymers or gradient blocks can be obtained by polymerizing monomers of different polarities and/or if one component is consumed at a faster rate as it was observed for the copolymerization of methyl methacrylate with butyl acrylate.³⁷ Generally, in radical homopolymerization reactions, acrylates were found to be more reactive than corresponding methacrylates.³⁸ Different reaction rates became apparent as higher monomer reactivity was found for styrene when copolymerized with acrylates than for the methacrylic analogues.³⁹

Beyond the choice of methacrylate versus acrylate, recent studies demonstrated that marginal changes in structural architecture such as the charge arrangement of sulfo- and sulfobetaine polymers had severe effects on the fouling properties when tested against proteins, diatoms, and zoospores of the green algae *U. linza*.^{40,41} Nonspecific protein adsorption was strongly affected when the carbon spacer between the carboxyl and ammonium groups in carboxybetaines was varied.^{42,43} Also, the polymer backbone plays a role in the resistance of polymers, as methacrylates and methacrylamides showed different inert properties.⁴⁰ While several reports already noted the high impact of subtle variations on the inert properties of acrylate coatings, a systematic study on the impact of the additional methyl group in methacrylates is still lacking. Although both acrylates and corresponding methacrylates have been included in several antifouling studies, the results are rarely discussed comparatively with respect to differences in fouling control. On temperature-responsive poly(butyl acrylate) (PBA) and methacrylate (PBMA), the protein adsorption of bovine serum albumin (BSA) appeared to be very similar on both components.⁴⁴ In the case of poly(2-hydroxyethyl acrylate) (PHEA) and methacrylate (PHEMA), the number of adhered platelets and protein⁴⁵ was lower on the methacrylate analogue, whereas no significant difference toward the settlement of barnacle cyprids⁴⁶ or cell adhesion⁴⁷ was observed.

Here, we address the question if the choice of methacrylate or acrylate monomers affects the inert properties of amphiphilic polymer coatings. The polymers were synthesized based on a recently discovered amphiphilic formulation consisting of 5 wt % 3-[[2-(methacryloyloxy)ethyl]-

dimethylammonio]propionate (CBMA) and 95 wt % DCPEA.³² The zwitterionic CBMA or carboxybetaine acrylate (CBA) were in situ copolymerized with DCPEA or ethylene glycol dicyclopentenyl ether methacrylate (DCPEMA) as a supporting hydrophobic matrix by grafting through using 3-(trimethoxysilyl)-propyl methacrylate (MPTMS) functionalization for a covalent attachment to the substrates. The resulting polymer coatings were characterized by surface analytics, and their AF and FR capabilities were tested against the model marine diatom *N. perminuta* in a dynamic accumulation experiment and against multiple species in a dynamic short-term field exposure (DSFE) assay.

MATERIALS AND METHODS

Chemicals. Zwitterionic 3-[[2-(methacryloyloxy)ethyl]-dimethylammonio]propionate (CBMA) was purchased from TCI. 2-(Dimethylamino)ethyl acrylate (DMAEA), ethylene glycol dicyclopentenyl ether acrylate (DCPEA), ethylene glycol dicyclopentenyl ether methacrylate (DCPEMA), ethylene glycol diacrylate (EGDA), 2,2-dimethoxy-1,2-diphenylethan-1-one (DMPAP), 3-(trimethoxysilyl)propyl methacrylate (MPTMS), and butyl methacrylate (BMA) were obtained from Sigma-Aldrich. Glutaraldehyde was acquired from Fisher Scientific. Ethanol, diethyl ether, toluene, and acetonitrile were purchased from VWR International. β -Propiolactone was obtained from ABCR. 2,2,2-Trifluoroethanol (TFE) was purchased from Carl Roth. All chemicals were used as received.

Synthesis of Carboxybetaine Acrylate (CBA). With slight modifications of a previously published protocol, CBA was synthesized by a ring-opening reaction of β -propiolactone with DMAEA.¹¹ In short, DMAEA (1.5 mL, 15.0 mmol) was dissolved in 60 mL of dried acetone under a nitrogen atmosphere. The solution was cooled down to 4 °C and β -propiolactone (1.0 mL, 15.9 mmol) was added dropwise to the solution over 30 min. The reaction mixture was further stirred at 4 °C for 8 h. A white solid precipitated and was centrifuged (Hettich Centrifuge Universal 320R) at 5000 rpm in light-sealed falcon tubes. The product was washed twice with anhydrous ether and subsequently dried under reduced pressure. The reaction yielded 14% of a white solid, which was stored at 4 °C under argon until further use. Nuclear magnetic resonance (NMR) was measured on a Bruker spectrometer in deuterated water. ¹H NMR (200 MHz, D₂O): δ 2.76 (t, 2H, CH₂COO), 3.19 (s, 6H, NCH₃), 3.68 (t, 2H, NCH₂), 3.79 (t, 2H, CH₂N), 4.65 (s, 2H, OCH₂), 6.06 (m, 1H, =CH), 6.22 (m, 1H, =CH), 6.43 (m, 1H, =CH).

Nuclear Magnetic Resonance. Nuclear magnetic resonance (NMR) measurements were performed on a Bruker Avance DPX-200 spectrometer at 303 K equipped with a Bruker B-AVS 60 sample changer. The software MestReNova 10.0.2 was used to analyze the NMR spectra. Chemical shifts δ were given in parts per million (ppm) against the internal standard. Following abbreviations were used: s (singlet), d (doublet), t (triplet), dd (doublet of doublet), dt (doublet of triplet), td (triplet of doublet), and m (multiplet).

Surface Preparation. Before the polymerization was carried out, smooth glass slides (Nexterion B, Schott, Germany) were prefunctionalized with 3-(trimethoxysilyl)propyl methacrylate (MPTMS) following previous reports.^{32,48} Briefly, the slides were plasma-treated with oxygen for 20 min (0.4 mbar, 80 W, 22 kHz, miniFlecto-PC-MFC, Gala Instruments, Germany) to render the surface hydrophilic. Subsequently, the activated slides were immersed in a degassed solution of 10% (v/v) MPTMS in toluene and sonicated for 16 h at 50 °C. Subsequently, the slides were gently rinsed with ethanol and dried in a stream of nitrogen before curing them in a vacuum oven at 85 °C for 30 min. Uncoupled MPTMS was removed by 20 s sonication in ethanol with subsequent drying in a stream of nitrogen gas. After drying the MPTMS-modified substrates, they were stored under argon or immediately used.

Preparation of the Polymer Coatings. The polymer coatings were produced by grafting through photoinitiated free radical

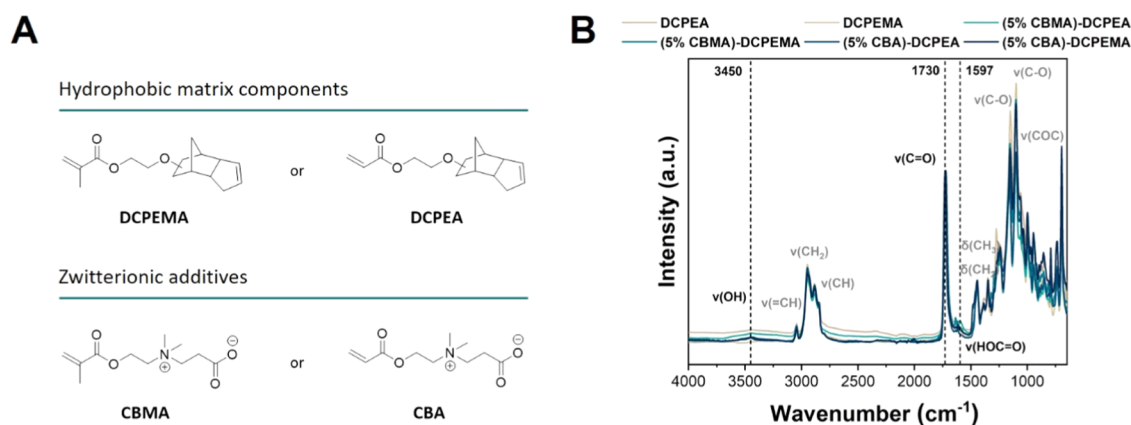


Figure 1. (A) Chemical structures of zwitterionic carboxybetaine methacrylate (CBMA) and the corresponding acrylate (CBA) as well as of ethylene glycol dicyclopentenyl ether methacrylate (DCPEMA) and the corresponding acrylate (DCPEA). (B) IR analysis of the different acrylic and methacrylic polymer coatings: the spectra were normalized to the carbonyl stretching vibration at 1730 cm^{-1} .

polymerization, following a previously reported procedure.³² The methacrylate group of MPMTS (tie coat) provides polymerizable binding sites for covalent attachment of the monomers to the surface. For the monomer solutions, 5 wt % of either CBMA or CBA and 95 wt % of either DCPEMA or DCPEA were dissolved in TFE (100 wt % solvent/monomer). DMPAP (DMPAP/monomer, 1 wt %) as the photoinitiator dissolved in acetonitrile and 3 wt % EGDA (EGDA/monomer) as the crosslinker were added. Subsequently, the light-protected monomer solution was degassed for 45 min. Coatings of the DCPEA, DCPEMA, and BMA homopolymers were prepared by dissolving 100 wt % hydrophobic monomer in TFE and adding the same amount of EGDA and DMPAP as before. The monomer mixtures were applied to a HybriWell reaction chamber (SecureSeal adhesive chamber, chamber volume: 275–320 μL , size L $\times$ W $\times$ thickness: 60 mm $\times$ 21 mm $\times$ 0.15 mm, ports diam.: 3.2 mm; Grace Bio-Labs) through sealable ports to perform the polymerization under controlled, airtight conditions. The polymerization was initiated by UV irradiation (UVAcube 100, Dr. Hönle AG, 100 W iron-doped mercury vapor lamp, cutoff filter $>310\text{ nm}$, source–sample distance: 18 cm, 15 mW/cm^2) between 5.5 and 7.5 min, depending on the monomers used (DCPEA (5.5 min), CBMA-DCPEA (6.0 min), CBA-DCPEA (6.0 min), DCPEMA (6.5 min), CBMA-DCPEMA (6.5 min), CBA-DCPEMA (6.5 min), and BMA (7.5 min)). After irradiation, the reaction vessel was kept sealed for 16 h to allow the monomer to diffuse and react. Subsequently, the sealing system was detached, and the resulting coatings were leached in TFE for 30 min to remove unpolymerized monomers. The slides were rinsed in water and dried in a stream of nitrogen before storing them under argon until further use.

Attenuated Total Reflection (ATR)-Fourier Transform Infrared (FTIR) Spectroscopy. ATR-FTIR spectra were obtained on a nitrogen-purged Vertex 70 spectrometer (Bruker, Ettlingen, Germany) equipped with a platinum diamond ATR crystal and an RT-DLaTGS detector. For the displayed spectra, 50 scans at a resolution of 4 cm^{-1} were recorded in a wavenumber range of $650\text{--}4000\text{ cm}^{-1}$.

Contact Angle Goniometry. Using two custom-built goniometers, both the static water contact angle (WCA) in air and the captive bubble contact angle (CBCA) under water were determined.³² For the static WCA, a droplet of tridistilled water (~ 10 to $15\text{ }\mu\text{L}$) was deposited on the polymer film and recorded via a charge-coupled device (CCD) camera to automatically fit the shape with Young's equation. For the CBCA, the samples were immersed in a transparent water vessel with the coatings facing downward. An air bubble (~ 10 to $15\text{ }\mu\text{L}$) was dispensed onto the coatings from below and recorded by a CCD camera and manually analyzed by applying the angle tool of the software ImageJ, following a previously published protocol.³² Each value represents the mean of three independent measurements, with error bars representing the standard deviation.

Atomic Force Microscopy (AFM). Topographic images ($20\text{ }\mu\text{m} \times 20\text{ }\mu\text{m}$) were measured on a NanoWizard 3 (JPK, Germany) equipped with a MULTI75GD-G (Budget Sensors, Bulgaria) and a force constant of 3 N/m. Typical settings for the measurement in air were a 2 V setpoint, 750 Hz i-gain, and a tip velocity of $89\text{ }\mu\text{m}/\text{s}$. To determine the surface roughness, images were plane-fitted and analyzed by the root-mean-square (RMS) roughness function of the software package Gwyddion (Version 2.53). The displayed values are the mean of three different areas measured on the height trace image. Error bars represent the standard deviation.

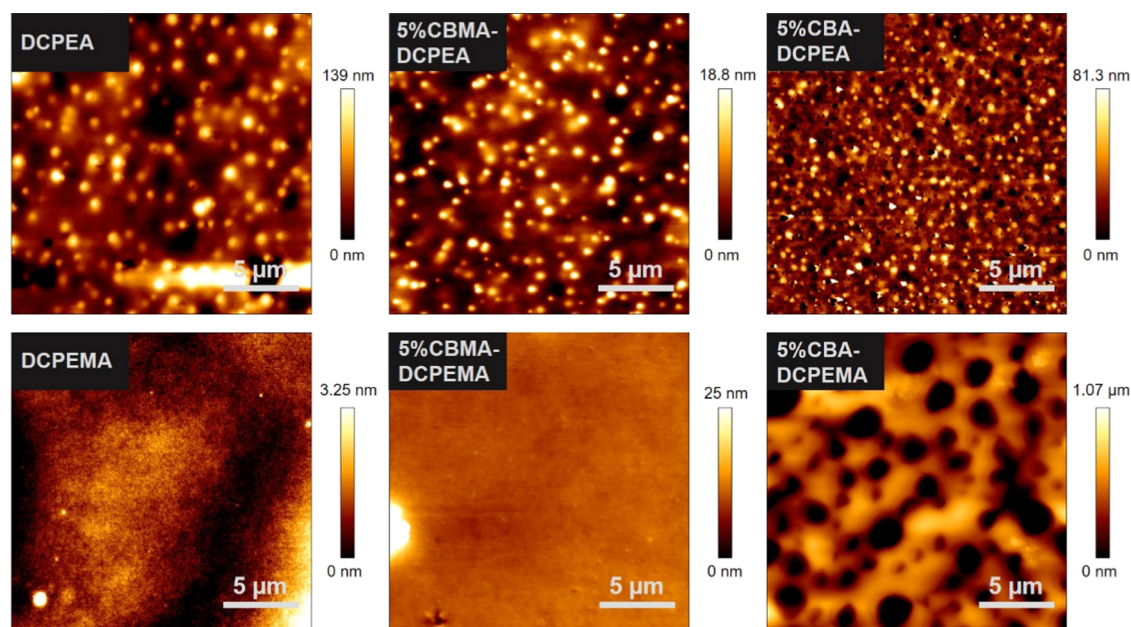
Dynamic Microfluidic Diatom Attachment Assay with *N. perminuta*. To evaluate the AF and FR behavior of the polymer coatings, a dynamic attachment assay with the diatom species *N. perminuta* was performed in a microfluidic accumulation experiment.⁴⁹ The cultivation of the diatoms was carried out according to a previously described protocol.⁴⁹ In brief, *N. perminuta* purchased from the Culture Collection of Marine Algae and Protozoa (Dunbeg, Great Britain) was received in sterilized nutrient-rich seawater (CaribSea, Fort Pierce, Florida) and adjusted to a pH of 8.1. The cultivation was performed in a light cabinet (Snijder Scientific, Tilburg, the Netherlands) with a day/night cycle of 12/12 h. For the assays, the diatoms were detached from the bottom of the culture flask by sonication and the culture medium was exchanged against nutrient-free seawater. Using the OD444 determined by a UV/vis spectrometer (Cary 60, Agilent Technologies, Waldbronn, Germany), the density of the diatom suspension was adjusted to a concentration of 2 million diatoms/mL. During the assay, the suspension was pumped through the microfluidic channels with a laminar flow profile and a wall shear stress of 0.16 Pa for 90 min. Prior to fluorescence microscopy analysis, the channels were rinsed with pure seawater for 10 min at the same settings to remove unattached diatoms.

Dynamic Short-Term Field Exposure (DSFE) Assay. The samples were further evaluated by subjecting the polymers to a dynamic short-term field exposure assay at the Port Canaveral test site of the Florida Institute of Technology. Being exposed to the real ocean environment, the coatings were challenged by a multispecies population. For the dynamic exposure, the samples were flush mounted to the surface of a custom-built rotating disk that exerted a comparable wall shear stress (0.18 Pa) on the coatings as the dynamic microfluidic laboratory experiment when operated at a rotation speed of 12 rpm.⁵⁰ The coatings were retrieved after 5 days of immersion, fixed with a 2.5 wt % glutaraldehyde solution for 10 min, washed twice in MQ water for 10 min each, and allowed to air dry.

Microscopy and Image Analysis. The density of biofouling organisms on the surfaces of both the dynamic microfluidic laboratory attachment assays and the dynamic field exposures were characterized by phase contrast and fluorescence microscopy. The coatings were automatically imaged using an Eclipse Ti microscope from Nikon with a 10 $\times$ phase-contrast objective (CFI Plan Fluor DLL NA 0.3,

Table 1. Coating Preparation Conditions, RMS Roughness, Water Contact Angle, and Captive Bubble Contact Angle of the Synthesized (Co)polymers in Air and under Water

polymers	irradiation time [min]	curing time [h]	RMS roughness [nm]	WCA [°]	CBCA [°]
DCPEA	5.5	16	24.1 ± 1.0	99 ± 4	50 ± 1
DCPEMA	6.5	16	0.6 ± 0.1	102 ± 5	50 ± 3
(5% CBMA)-DCPEA	6.0	16	4.3 ± 0.2	92 ± 3	17 ± 2
(5% CBMA)-DCPEMA	6.5	16	3.8 ± 1.9	96 ± 5	25 ± 3
(5% CBA)-DCPEA	6.0	16	18.5 ± 1.2	85 ± 3	30 ± 2
(5% CBA)-DCPEMA	6.5	16	234.4 ± 10.9	88 ± 3	37 ± 2

**Figure 2.** Representative AFM height images (20 $\mu\text{m} \times 20 \mu\text{m}$) showing the morphology of the polymer coatings of DCPEA, DCPEMA, (5% CBMA)-DCPEA, (5% CBA)-DCPEA, (5% CBMA)-DCPEMA, and (5% CBA)-DCPEMA. The magnitude of the corrugation is provided by the colorbars.

Nikon, Germany) and a BV-2A fluorescence filter set (excitation: 420 nm; dichroic mirror: 455 nm; barrier filter: 470 nm, Nikon Instruments) along a predefined grid of 30 fields of view (dynamic attachment assay) and 60 fields of view (DSFE assay). The quantification of the degree of fouling was done using the software package NIS Elements (AR 4.30.02, Nikon, Germany). The density of fouling organisms in the laboratory assay was determined by automated counting of the attached diatoms, and the coverage after the dynamic field exposure was analyzed as fluorescence coverage using the autofluorescence of the attached marine organisms.

RESULTS AND DISCUSSION

Surface Characterization. The acrylate and methacrylate monomers (CBA, CBMA, DCPEA, and DCPEMA, see Figure 1A) were in situ (co)polymerized by photoinduced grafting through free radical polymerization in the direct vicinity of substrates premodified with methacrylate binding sites for covalent coupling, following a previously described procedure.³² The different monomers were dissolved in 2,2,2-trifluoroethanol (TFE) to overcome immiscibility between zwitterionic (CBA and CBMA) and the hydrophobic compounds (DCPEA and DCPEMA). The photoinitiator (DMPAP) as well as the crosslinker (EGDA) were added to the solution, and the polymerization under controlled, oxygen-free conditions was initiated by UV irradiation. The irradiation times were slightly varied (Table 1), as different initiation times were required for the acrylate and methacrylate

monomers. The obtained polymers are denoted as BMA (100 wt % BMA), DCPEA (100 wt % DCPEA), DCPEMA (100 wt % DCPEMA), (5% CBMA)-DCPEA (5 wt % CBMA and 95 wt % DCPEA), (5% CBMA)-DCPEMA (5 wt % CBMA and 95 wt % DCPEMA), (5% CBA)-DCPEA (5 wt % CBA and 95 wt % DCPEA), and (5% CBA)-DCPEMA (5 wt % CBA and 95 wt % DCPEMA). The general stability of these ethylene glycol dicyclopentenyl- and carboxybetaine-based coatings was characterized in a previous publication³² and therefore not repeated here. The chemistry of the polymer coatings was analyzed by ATR-FTIR spectroscopy (Figure 1B). All spectra showed typical CH stretching and bending vibrations, i.e., $\nu(\text{=CH})$ at 3045 cm^{-1} , $\nu(\text{CH}_2)$ at 2947 cm^{-1} , $\nu(\text{CH})$ at 2885 cm^{-1} , $\delta(\text{CH}_2)$ at 1446 cm^{-1} , and $\delta(\text{CH})$ at 1385 cm^{-1} . Also, the presence of a strong ester peak at 1730 cm^{-1} ($\nu(\text{C=O})$), the $\nu(\text{C-O})$ acyl and alkoxy stretching vibrations at 1150 and 1097 cm^{-1} , respectively, and the ether absorption band $\nu(\text{COC})$ at 1060 cm^{-1} are in good agreement with the molecular structures of both the hydrophobic DCPEA/DCPEMA and the zwitterionic CBA/CBMA compounds. The weakly pronounced peak at 1484 cm^{-1} from the CH_3 bending vibration allows to distinguish the DCPEMA-based polymers from the acrylic DCPEA analogues. Although the spectra of the amphiphilic polymers show great similarity to those of the homopolymers, the successful embedment of CBA/CBMA into the hydrophobic matrix can be observed in

the less pronounced but zwitterion characteristic broad band $\nu(\text{OH})$ at 3450 cm^{-1} and small carboxyl stretching vibration $\nu(\text{HOC}=\text{O})$ at 1597 cm^{-1} . The wettability of the polymers was determined by static water contact angle (WCA) goniometry (Table 1). The purely hydrophobic coatings DCPEA ($99 \pm 4^\circ$) and DCPEMA ($102 \pm 5^\circ$) showed unsurprisingly high WCA values. DCPEMA, carrying additional methyl groups in the alkyl backbone, was slightly more hydrophobic than DCPEA. For the amphiphilic polymer formulations, the presence of the zwitterionic components CBA and CBMA led to more hydrophilic WCAs than the purely hydrophobic coatings. In the case of CBMA, the variation of the hydrophobic matrix (DCPEA or DCPEMA) led to higher WCAs for (5% CBMA)-DCPEMA ($96 \pm 5^\circ$) compared to (5% CBMA)-DCPEA ($92 \pm 3^\circ$). (5% CBA)-DCPEMA exhibited a WCA of $88 \pm 3^\circ$, whereas for (5% CBA)-DCPEA, a WCA of $85 \pm 3^\circ$ was measured. Thus, DCPEA was more hydrophilic than DCPEMA, and zwitterionic acrylate CBA lowered the contact angle of the amphiphilic coating as compared to methacrylate CBMA.

The surface morphology and the RMS roughness of the coatings were determined by AFM (Table 1). DCPEMA revealed a very smooth ($0.6 \pm 0.1\text{ nm}$) and featureless surface, while DCPEA was much rougher ($24.1 \pm 1.0\text{ nm}$) and a more pronounced surface morphology with several localized protrusions was detected (Figure 2). All amphiphilic CBMA-containing coatings showed lower RMS roughness values than the CBA-containing polymers; (5% CBMA)-DCPEA had a roughness of $4.3 \pm 0.2\text{ nm}$, whereas for the methacrylate-methacrylate polymer (5% CBMA)-DCPEMA, a roughness of $3.8 \pm 1.9\text{ nm}$ was found. On the contrary, the amphiphilic polymers (5% CBA)-DCPEA and (5% CBA)-DCPEMA containing zwitterionic acrylate exhibited surface roughnesses of $18.5 \pm 1.2\text{ nm}$ (DCPEA matrix) and $234.4 \pm 10.9\text{ nm}$ (DCPEMA matrix). The surface morphology found on DCPEA remained similar when copolymerized with either CBMA or CBA with slightly smoother structures on (5% CBA)-DCPEA than on (5% CBMA)-DCPEA. For DCPEMA, the integration of CBMA into the matrix led to no obvious changes compared to the homopolymer, while CBA led to a slightly more porous structure with noticeable holes.

Besides the above-presented wettability analysis by static water contact angle goniometry in air, the underwater rearrangement behavior of amphiphilic polymers can be analyzed by captive bubble contact angle goniometry (CBCA).³² It was observed that the presence of zwitterions in the polymers led to much lower contact angles (Figure 3) compared to the values measured by sessile droplet static WCA goniometry. The WCAs in air were $>85^\circ$ and thus much more hydrophobic, even in the case of the amphiphilic polymers. In the underwater CBCA measurements, the amphiphilic coatings turned more hydrophilic and CBCAs of 18° for (5% CBMA)-DCPEA, 25° for (5% CBMA)-DCPEMA, 30° for (5% CBA)-DCPEA, and 37° for (5% CBA)-DCPEMA were found. In the CBMA-containing films, a decrease of the contact angle by 74° (DCPEA copolymer, a WCA of 92° in air) and 71° (DCPEMA copolymer, a WCA of 96° in air) was observed. The CBA-containing polymers, which revealed higher hydrophilicity in air than their CBMA analogues, turned out to be less hydrophilic in the CBCA measurements. The decrease in the contact angle under water was 55° (DCPEA copolymer, a WCA of 85° in air) and 51° (DCPEMA copolymer, a WCA of 88° in air). We noted that the contact

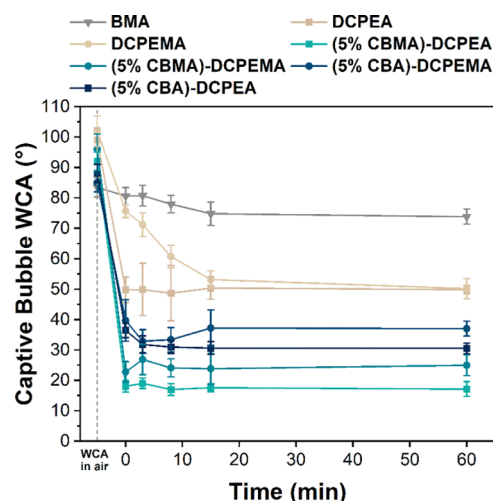


Figure 3. Captive bubble contact angles measured under water on polymer films immersed for 0, 3, 8, 15, and 60 min. For the purpose of illustration, the WCAs in air (Table 1) were included in the graph.

angles equilibrated within only a few minutes on all amphiphilic polymers, indicating a quick reorientation under water. A decrease in hydrophobicity was also observed for the homopolymers DCPEA and DCPEMA. Consequently, DCPEA that exhibited a WCA in air of 99° showed a CBCA of 50° . On DCPEMA with a WCA of 102° , an equilibrated CBCA of 50° was obtained. Interestingly, the rearrangement kinetics was very fast for DCPEA and the amphiphilic polymers, which equilibrated within minutes. On the contrary, the methacrylate DCPEMA required more than 10 min to change the CBCA from 76 to 53° , which then seemed to slowly equilibrate and finally reached a value of 50° after 60 min. The final wettability of the DCPEMA and DCPEA copolymers was thus very similar.

Dynamic Microfluidic Diatom Attachment Assay with *N. perminuta*. The polymers were tested in two different biological assays under dynamic conditions (Figure 4). In the laboratory assay, the coatings were subjected to a dynamic, microfluidic attachment assay against the diatom *N. perminuta* in a custom-built microfluidic setup.⁴⁹ The experiment was conducted for 90 min at a wall shear stress of 0.16 Pa . Hydrophobic BMA was included in the assay as a negative reference and the diatom density on all other samples was normalized to BMA. It can be seen from the results of the assay in Figure 4A that relative to BMA, the hydrophobic coatings DCPEA and DCPEMA reduced the settlement of diatoms by 45% on DCPEA and even by 78% on DCPEMA. The resistance toward diatom attachment was further enhanced in the case of the amphiphilic coatings when zwitterionic CBA or CBMA was incorporated into the hydrophobic DCPEA or DCPEMA matrix. In the case of DCPEA, the employment of CBMA led to a relative diatom adhesion of 0.01 (99% lower than on BMA), while 5% CBA-DCPEA resulted in a slightly higher relative diatom adhesion of 0.05, which still implies a 95% lower attachment compared to BMA. This improvement of the FR properties was also, but to a lesser extent, observed when the zwitterionic acrylate CBA or methacrylate CBMA was embedded into DCPEMA. (5% CBMA)-DCPEMA yielded a relative adhesion of 0.08 (92% lower than BMA), and (5% CBA)-DCPEMA reduced diatom densities to 0.12 as compared to BMA (88% reduction). Both CBMA-containing coatings revealed lower attachment than the CBA analogues.

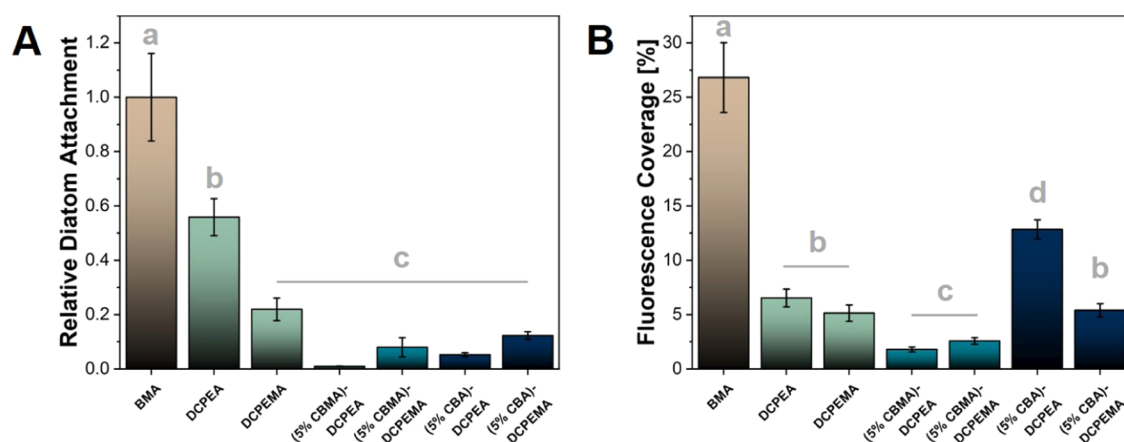


Figure 4. Biofilm formation on the acrylate/methacrylate polymer coatings. (A) Densities of *N. perminuta* diatoms on the polymer films relative to the hydrophobic control BMA. The coatings were subjected to a dynamic microfluidic attachment assay for 90 min. The flow created a wall shear stress of 0.16 Pa. (B) Fluorescence coverage after immersing the polymer coatings for 5 days in the ocean at the Port Canaveral test site of the Florida Institute of Technology. The samples were mounted on a rotating disk operated at a rotation speed of 12 rpm to generate a wall shear stress of 0.18 Pa. Each value represents the average of three measured replicates on which 30 fields of view (A) and 60 fields of view (B) were analyzed. Error bars represent the standard error of the mean. The statistical significance (ANOVA with post-hoc Tukey test, $p < 0.05$) of differences between the bars is provided as lowercase characters (a–d) and polymers with different labels show significant differences.

As indicated by the letters in Figure 4A, a statistical test (analysis of variance (ANOVA) with post-hoc Tukey) showed that the reduction in diatom density on DCPEMA and all amphiphilic coatings was statistically significantly lower than those on BMA and DCPEA ($p < 0.05$). The differences among the amphiphilic polymers and compared to DCPEMA, however, were not statistically significant ($p > 0.05$).

Dynamic Short-Term Field Exposure (DSFE) Assay. To demonstrate the relevance of the observed effects in a real ocean scenario, the polymers were immersed under dynamic conditions at Port Canaveral using a rotating disk setup, which was operated at a rotation speed of 12 rpm to generate a wall shear stress of 0.18 Pa, for five days.⁵⁰ After retrieval of the samples, the degree of fouling was assessed by determining the biomass-covered surface area by fluorescence microscopy using the autofluorescence of the marine organisms. As shown in Figure 4B, the highest accumulation with 27% coverage was found on BMA, which was reduced by 74% on DCPEA (7% coverage) and by 81% on DCPEMA (5% coverage). The integration of zwitterionic moieties into the DCPEA matrix considerably changed the biofilm attachment. In the case of CBA, the polymer (5% CBA)-DCPEA revealed a coverage of 13%, which was almost twice as high as pure DCPEA and the second highest fouling level after BMA. CBMA incorporation into (5% CBMA)-DCPEA in turn led to high fouling resistance with a coverage of <2%, i.e., a reduction by 93% compared to BMA. For methacrylate DCPEMA (5% coverage), the copolymerization with the zwitterionic acrylate CBA did not improve the fouling properties and a coverage of 5% was detected on the amphiphilic polymer (5% CBA)-DCPEMA. CBMA, on the other hand, strongly improved the fouling resistance of DCPEMA, and with a coverage of only 2.5%, biofilm formation was reduced by 50% compared to DCPEMA and by 91% compared to BMA. Different labels in Figure 4B indicate statistically significant (ANOVA with post-hoc Tukey, $p < 0.05$) differences between the coatings. BMA and (5% CBA)-DCPEA showed statistically significantly higher fouling ($p < 0.05$) than all other coatings and were different from each other. The amphiphilic CBMA-containing polymers (5% CBMA)-DCPEA and (5% CBMA)-DCPEMA performed

significantly better ($p < 0.05$) than all of the other tested polymers.

In summary, a set of hydrophobic and amphiphilic acrylate- and methacrylate-based polymers were synthesized using grafting through photoinitiated free radical polymerization. The hydrophobic monomers DCPEA and DCPEMA were blended with 5 wt % zwitterionic CBA or CBMA and in situ copolymerized in the presence of a crosslinker (3 wt %) and a photoinitiator (1 wt %) while simultaneously being covalently anchored to a methacrylate–silane-functionalized substrate. The successful incorporation of zwitterions into the hydrophobic matrix was supported by the zwitterion characteristic broad OH absorption band at 3450 cm^{-1} and the carboxyl stretching vibration at 1597 cm^{-1} in the ATR-FTIR spectra. The sufficient stability of ethylene glycol dicyclopentenyl ether and carboxybetaine copolymer coatings in SW was verified in a previous study.³² The incorporation of CBMA and CBA into the DCP matrix led to a static WCA decrease from 99° for DCPEA to 88° for (5% CBMA)-DCPEA and to 85° for (5% CBA)-DCPEA. Also, the slightly more hydrophobic DCPEMA (102°) showed a reduced hydrophobicity for the amphiphilic CBMA (96°) and CBA (92°) copolymers. In all cases, the methacrylates were more hydrophobic than the acrylates, which is a result of the additional methyl group in the methacrylates and thus the overall amount of hydrophilic atoms in the film, i.e., oxygen, is reduced. Since the ability to form stable hydration layers under water is proposed as the key driving force of the ultralow fouling properties of zwitterions,^{51,52} the capability of the polymers to reorganize to expose the hydrophilic groups at the interface was evaluated using CBMA. A strong shift to more hydrophilic contact angles became apparent. The enhanced exposure of hydrophilic groups by a rearrangement is in accordance with previous reports.³² Even the purely hydrophobic acrylate DCPEA (50°) and the methacrylate DCPEMA (76°) showed higher hydrophilicities within only a minute of immersion in water. While all four amphiphilic polymers and DCPEA rapidly rearranged within minutes, DCPEMA took much longer. The fast rearrangement kinetics of the first five polymers can be explained by a simple rotation along the backbone (crankshaft

motion).⁵³ The slower rearrangement of DCPEMA indicates either the inhibition of this rotation or a rearrangement within the polymer.

An analysis of the polymer morphology was conducted by AFM. The DCPEA and DCPEMA homopolymers had different morphologies, which was also reflected in the surface roughness, expressed in a very smooth (RMS: 0.6 nm) featureless surface of DCPEMA, while DCPEA was slightly rougher (RMS 24.1 nm), which can possibly be caused by a solvent effect that was already previously observed in the literature.⁴⁰ When the two materials were copolymerized with carboxybetaine methacrylate, roughnesses of 4.3 nm for (5% CBMA)-DCPEA and 3.8 nm for (5% CBMA)-DCPEMA were found. When the zwitterionic acrylate CBA was copolymerized with the methacrylate matrix DCPEMA, a rather inhomogeneous surface with a comparably high surface roughness (234.4 nm) was obtained. If CBA was incorporated into the acrylate DCPEA, the surface roughness of 18.5 nm and the surface corrugation were found to be similar to the homopolymer.

The slowest reorganization of CBCA was found for the DCPEMA homopolymer, which was at the same time the coating with the smoothest morphology and the smallest RMS value of 0.6 nm. While the RMS values of most of the other polymers were in a similar ballpark, (5% CBA)-DCPEMA stuck out due to very high roughness. At the same time, the morphology strongly differed, as the surface was not characterized by protrusions but by holes and the interface appeared to be more porous. Among the possible reasons for a different morphology of the copolymers could be the different reactivity of the acrylates and the methacrylates. In free radical polymerization synthesis, the reactivity of the acrylates was found to be higher than that of the corresponding methacrylates.³⁸ While the methacrylate/acrylate (5% CBMA)-DCPEA was found to be rather smooth, the acrylate/methacrylate copolymer (5% CBA)-DCPEMA showed a rougher morphology. One possible explanation is that the higher reactivity of CBA, a hydrophilic compound, is likely to form hydrophilic-enriched segments in the polymer, while the polymerization of the less reactive DCPEMA requires more time to be completed. The situation is reversed if the hydrophilic CBMA reacts with the more reactive DCPEA, as first a network is created, and then with diminishing DCPEA monomer concentration, CBMA is increasingly incorporated into the polymer. Here, it became apparent that the copolymers based on the same polymerizable functional group, i.e., methacrylate/methacrylate and acrylate/acrylate, provided more homogeneous morphologies as opposed to more heterogeneous looking structures of acrylate/methacrylate formulations. The hypothesized presence of CBA-enriched segments in the amphiphilic network is further supported by the AFM phase images (Figure S1). While the less reactive methacrylate CBMA resulted in surfaces with small overall phase variations ($\leq 4^\circ$, $\text{RMS} \leq 0.6^\circ$), CBA-containing polymer coatings showed much stronger phase variances ($>100^\circ$, $\text{RMS} > 27.1^\circ$), regardless of whether acrylic DCPEA or methacrylic DCPEMA was used as the hydrophobic polymer matrix. Thus, despite the distinct morphologies, the phase data of the CBMA-based polymer blends revealed small local differences in physiochemical properties and hence similar local chemistries, whereas the much higher roughness of the CBA analogues was accompanied by high phase variations, which confirms a strongly different local chemistry. This is in line with the before-mentioned suggestion

that the higher reactivity of CBA compared to CBMA causes the formation of segments rich in hydrophilic building blocks.

When evaluated against the attachment of the model diatom *N. perminuta* in the dynamic microfluidic laboratory attachment assay and against multiple species in field exposures, strong differences in the fouling repellency of acrylate- and methacrylate-based polymers were observed. In both assays, the hydrophobic methacrylate DCPEMA and acrylate DCPEA showed severely reduced accumulation of organisms compared to the negative reference BMA. The methacrylate DCPEMA showed even further reduction of the diatom accumulation (by 78%) compared to DCPEA (by 45%). This trend was also obtained in the field, where DCPEA reduced fouling by 74% (7% coverage) and DCPEMA by 81% (5% coverage). The slightly higher fouling on the acrylate was statistically not significant ($p > 0.05$). Despite their hydrophobic properties, DCPEA and DCPEMA exhibited very good fouling resistance, which is in line with previous reports about their reduction of three pathogenic freshwater bacteria¹⁴ and also against marine organisms.³² Since both homopolymers DCPEA and DCPEMA revealed equal contact angles after 1 h in the CBCA measurement, the better performance of DCPEMA might not be a result of wettability but ascribed to the lower surface roughness.

The copolymerization of DCPEA and DCPEMA with the zwitterionic monomers CBMA and CBA into amphiphilic networks enhanced the ability to resist diatoms in laboratory assays compared to the hydrophobic homopolymers by more than 85% compared to BMA. Each CBMA-containing amphiphilic polymer (5% CBMA)-DCPEA (99% reduction) and (5% CBMA)-DCPEMA (92% reduction) performed far better than the CBA counterparts (5% CBA)-DCPEA (95% reduction) and (5% CBA)-DCPEA (88% reduction). In the short-term dynamic field tests, again all tested polymers performed statistically significantly better ($p < 0.05$) than the BMA controls, as they reduce fouling by more than 50%. The hydrophobic homopolymers were performing very well (74% and 81% fouling reductions on DCPEA and DCPEMA, respectively, compared to BMA). The CBMA-containing copolymers performed best with a 93% reduction on (5% CBMA)-DCPEA and a 91% reduction on (5% CBMA)-DCPEMA. While (5% CBA)-DCPEMA with an 81% reduction performed on a similar level as DCPEMA, the purely acrylic copolymer (5% CBA)-DCPEA performed significantly worse ($p < 0.05$), even than the hydrophobic homopolymers (52% reduction).

Yet, in particular, the copolymers based on CBMA in combination with DCPEA/DCPEMA performed very well. This good performance of the amphiphilic coatings is in line with the previous literature that demonstrated the high efficiency of amphiphilic zwitterionic polymers toward cell adhesion,⁵⁴ attachment of *N. incerta*⁵⁵ and multiple marine organisms.³² Despite small deviations in absolute values between CBMA- and CBA-containing films, the amphiphilic polymers showed no statistically significant differences ($p > 0.05$) among each other in the lab assay, as the overall diatom density was extremely low (<0.12) relative to BMA. In the field test, on the contrary, the FR properties varied conspicuously between acrylates and methacrylates. Several literature reports describe that the surface morphology is known to influence the fouling resistance of surfaces with higher roughness commonly exerting detrimental effects.^{56–58} Both CBA-containing amphiphilic coatings (5% CBA)-DCPEA and (5% CBA)-

DCPEMA revealed higher surface roughnesses than the amphiphilic CBMA coatings, which could contribute to their poorer performance in the field tests (52 and 92%), though the highest roughness of 234.4 nm of (5% CBA)-DCPEMA did not lead to the highest fouling. In this context, the phase images of the different amphiphilic polymers were found to be useful as the coatings based on CBMA were chemically more homogeneous than the CBA analogues. This together with the higher surface roughness could have caused the weaker performance of CBA in the field, while the impact of surface homogeneity on solely the diatom attachment in the laboratory environment seemed rather neglectable. These differences in physiochemical properties from the acrylic zwitterion to the methacrylic analogues could also explain the observed variances in CBCA, where the amphiphilic CBA polymers were surprisingly less hydrophilic than the methacrylic CBMA polymers, as a chemical heterogeneity is likely to influence the surface wettability.

In both laboratory and field tests, a generally better performance of methacrylates in comparison to acrylates was found, which was evident for the hydrophobic component DCPEMA both as a homopolymer and an amphiphilic polymer, and also for the zwitterionic component CBMA. It was interesting to observe that only minor changes such as an additional methyl group showed a high influence on the fouling resistance. This supports previous observations that marginal variations in structural architecture, such as charge arrangement in sulfo- and sulfobetaine polymers^{40,41} or the type of carbon spacer between carboxylate and amine moieties in carboxybetaines,⁴² can severely alter the antifouling properties. The well-performing CBMA showed slightly larger WCAs in air, which is counterintuitive to their performance in the first place, though CBCA measurements revealed the major rearrangement under water, which ultimately leads to a hydrophilic interface possibly responsible for lower fouling properties than the CBA analogues.

CONCLUSIONS

Hydrophobic ethylene glycol dicyclopentenyl ether acrylate and methacrylate were copolymerized into amphiphilic polymers using hydrophilic carboxybetaine acrylate or methacrylate by grafting through free radical polymerization. Then, the acrylic and methacrylic monomers were in situ (co)polymerized and covalently tethered to methacrylate-functionalized surfaces. A set of heterogeneous acrylate/methacrylate and homogeneous (meth)acrylate/(meth)acrylate compositions were characterized by AFM and by static water and captive bubble contact angle goniometry. The dynamic microfluidic attachment assay with *N. perminuta* and the short-term dynamic rotating disk assay with multiple organisms in the field revealed very good fouling-release properties of ethylene glycol dicyclopentenyl ether (meth)acrylate and all tested amphiphilic polymer formulations compared to hydrophobic butyl methacrylate. We demonstrated that even marginal changes in the monomer structure such as the additional methyl groups of methacrylates in contrast to acrylates strongly impacted the morphological structure and the resistance toward marine biofouling, in which the methacrylate homopolymers showed lower fouling properties than the acrylates. Furthermore, the choice of the incorporated acrylic or methacrylic zwitterion in the amphiphilic polymers decided if their FR performance was enhanced compared to the hydrophobic homopolymers during

field exposure. Here, carboxybetaine methacrylate clearly outperformed its acrylic analogue. Thus, this study showed that not only the functional moieties but also the polymerizable methacrylate or acrylate group of the monomers have to be carefully selected to fine-tune the FR properties and must be considered when designing effective polymer coatings for marine applications, especially if they are copolymerized into amphiphilic networks.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.1c00428>.

Representative phase images of dry polymer coatings determined by AFM analysis (PDF)

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Funding

Financial support from the Office of Naval Research (ONR), Grant Nos. N00014-20-12244 and N00014-20-1-2243, and from the DFG, Grant No. RO 2524/4-2, is gratefully acknowledged.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors would like to acknowledge helpful discussions with Prof. A. Laschewsky, University Potsdam. The authors further acknowledge access to the atomic force microscope by Prof. W. Schuhmann, Ruhr University Bochum.

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